

$$x_{spv} = [f_1, f_2, \ldots, f_{22}]^T, \;\; m \in \{0, 1\} \;\;^{22}$$

$$\mathcal{L}(\theta) = - \sum_{i=1}^N [w_{pos} y_i \ln \hat{p}_i + (1 - y_i) \ln (1 - \hat{p}_i)] + \Omega(\theta)$$

$$\hat{P}(Y = 1 \, | \, z) = \sigma(Az + B) = \frac{1}{1 + \exp(-(Az + B))}$$

$$f(x) = \phi_0 + \sum_{j=1}^{22} \phi_j(x)$$

$$c^* = \argmin_c \; {}_c \mathit{loss}(f(c), y^*) + \frac{\lambda_1}{k} \sum_{j \in \mathcal{M}} \frac{|c_j - x_j|}{MAD_j} + \lambda_2 \mathit{dpp}(c)$$

$$c_j = x_j \; \forall j \in \mathcal{I}, \;\; ||\Delta x||_0 \leq 3.0, \;\; c_j \in [0.0, 1.0] \; \forall j \in \mathcal{M}$$

$$S_{fused} = 0.35 \cdot S_{aud} + 0.35 \cdot S_{vid} + 0.30 \cdot S_{spk}$$

$$ECE = \sum_{m=1}^M \frac{|B_m|}{N} |acc(B_m) - conf(B_m)|$$